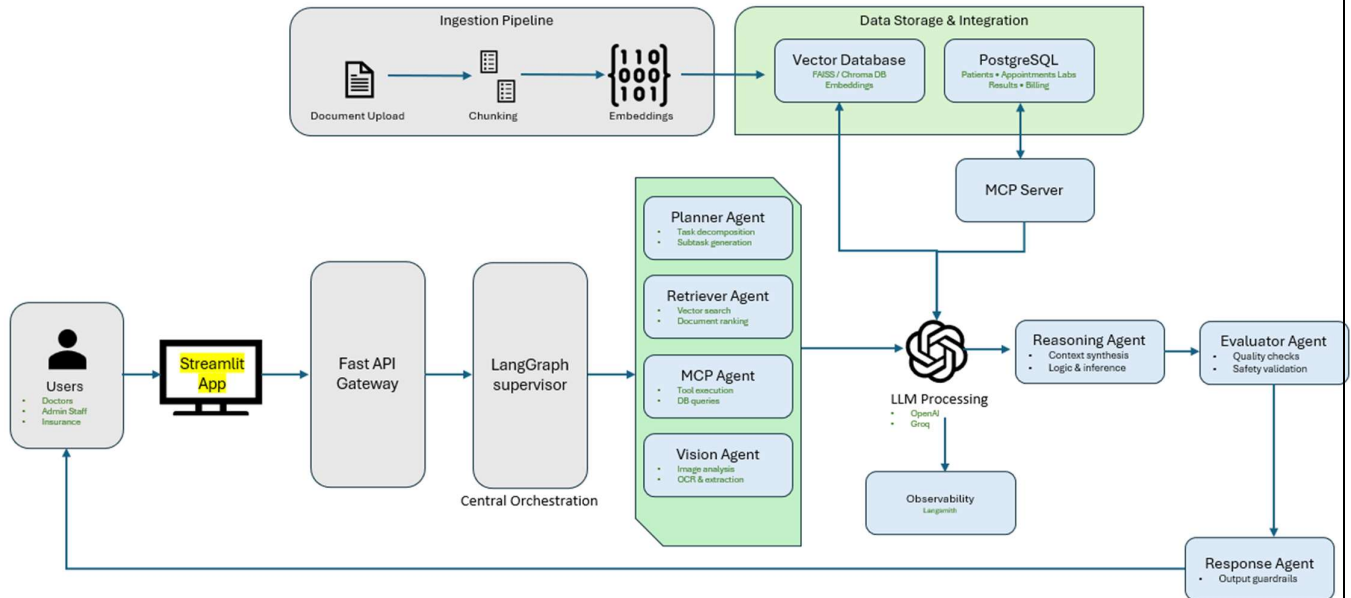


Enterprise Architecture for MediAssist AI



Architecture Breakdown

Users

Healthcare stakeholders such as Doctors, administrators, insurance teams, and hospital staff use the system to get information and perform tasks. They can ask questions, upload reports, and receive answers in simple language.

Streamlit Application

Streamlit is the user interface of the system. It allows users to chat with the AI, upload documents or images, and view results.

FastAPI Gateway

FastAPI works as the backend connection between the user interface and AI services. It receives requests from users and sends them to the correct components for processing. It handles request validation, authentication, session management, and error handling.

LangGraph Supervisor

The LangGraph Supervisor controls the overall workflow of the system. It decides which agent should perform each task and manages the flow of information between agents.

Planner Agent

The Planner Agent understands the user's request and breaks it into smaller tasks. It decides which agents, tools, and data sources are needed to answer the query.

Retriever Agent

The Retriever Agent searches the vector database for relevant documents and information. It helps the AI find the right knowledge before generating a response.

MCP Agent

The MCP Agent retrieves structured information from enterprise systems through MCP tools. It can access patient records, appointments, billing details, and lab results securely.

Vision Agent

The Vision Agent analyzes uploaded images such as prescriptions and medical reports. It extracts useful information like medicines, test results, and diagnoses from images.

Data Ingestion Pipeline

The Data Ingestion Pipeline prepares documents for the AI system. It processes uploaded files and converts them into a format that can be searched later.

Document Upload

This component allows healthcare documents such as SOPs, policies, reports, and guidelines to be uploaded into the system. These documents become part of the knowledge base.

Chunking

Chunking splits large documents into smaller sections. This makes searching faster and helps the AI retrieve more accurate information.

Embeddings

Embeddings convert text into numerical vectors that represent its meaning. These vectors help the system find similar information during searches.

Vector Database (FAISS / ChromaDB)

The Vector Database stores document embeddings for semantic search. It helps the system quickly find the most relevant document content for a user's question.

PostgreSQL Database

PostgreSQL stores structured hospital data such as patients, appointments, admissions, billing records, and lab results. It acts as the main enterprise database.

MCP Server

The MCP Server provides a secure connection between AI agents and the PostgreSQL database. It exposes database functions as tools that agents can use safely.

LLM Processing

The LLM Processing layer is the main AI engine of the system. It combines information from documents, databases, images, and user queries to generate answers.

Reasoning Agent

The Reasoning Agent analyzes all the collected information and generates meaningful responses. It creates summaries, recommendations, insights, and answers.

Evaluator Agent

The Evaluator Agent checks the quality of the AI-generated response. It verifies accuracy, relevance, grounding, and detects possible hallucinations before the response is shown to the user.

Response Agent

The Response Agent prepares the final response for the user. It applies safety checks, formats the output, and ensures the response is clear and user-friendly.

Observability (LangSmith)

The Observability layer monitors how the system is working. It tracks agent activities, tool usage, response times, errors, and overall system performance.

Data Storage & Integration Layer

This layer stores and manages both structured and unstructured data. It connects the vector database, PostgreSQL database, MCP server, and AI agents into one system.

Enterprise Knowledge Base

The Enterprise Knowledge Base contains hospital SOPs, policies, guidelines, compliance documents, and operational manuals. It provides trusted information for answering user questions.

End-to-End Workflow

The user submits a query through the application, which is routed through the FastAPI Gateway and LangGraph Supervisor. Relevant information is gathered by specialized agents, processed by the LLM, validated for accuracy and safety, and then returned to the user as a reliable and well-grounded response